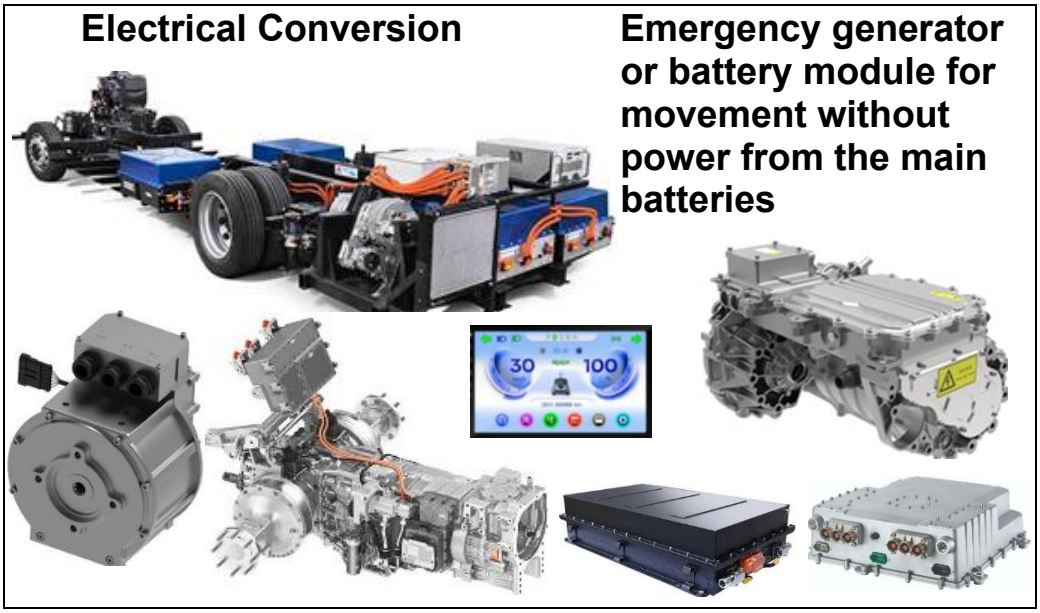
https://ecoes.eu/electric-conversion/

ECOES Electric Conversion

ECOES JV offers the conversion of new and used diesel-powered trucks, buses, and other mobile machinery to electric power.



What is ECOES electric conversion?

The ECOES electric conversion is a complex engineering process involving the replacement of the diesel engine with a liquid-cooled PMSM electric motor paired with an automatic transmission, as well as the modification of the steering, braking, electrical, hydraulic, and air conditioning systems, the drivetrain, the PTO, and all other systems related to the vehicle's functionality and safety.

The ECOES electric conversion process for new transport vehicles and other machinery does **not involve manufacturing new machines** from scratch. Instead, it utilizes new or used, mass-produced diesel-powered units—which have proven their reliability and efficiency in real-world operation—**and replaces their diesel engines with electric ones.**

2. Emergency module for movement without power from the main batteries

Any mobile machine that has undergone an ECOES electric conversion can optionally be equipped with an emergency module for movement without power from the main batteries. There are two types of emergency modules: generator-based and battery-based.

- **The emergency generator module** consists of a specially designed 400V AC generator that, via software and a charging unit, powers the vehicle's batteries to enable emergency operation and travel to a charging station.
- **The emergency battery module** consists of a separate battery that is not connected to the main battery pack used to drive the electric vehicle (such as a truck, bus, or other). When the

energy in the main battery pack is depleted, the driver switches the vehicle to the backup battery, which provides sufficient range to reach a charging station.

An emergency drive module that operates independently of the main batteries solves the most serious problem facing electric trucks:

- **What happens when the main battery runs out of power?**
- **ECOES electric trucks keep moving!**

3. Benefits of the ECOES electric conversion

3.1. Investment benefits:

* Prices for the ECOES electric conversion of new European diesel trucks into electric trucks are **30%–40% lower** than those of their electric counterparts.

* The ECOES electric conversion of existing used trucks extends the vehicle's service life by another **15–20 years** at a cost **50%–60% lower** than that of a new, comparable electric truck. Trucks—whether new or used—that undergo the ECOES electric conversion come with a standard two-year warranty, with the option to extend it.

3.2. Operational financial benefits:

* Charging costs for electric power are approximately **3–3.5 times (300–350%) lower** than diesel costs. According to various international studies, around **20–30%** of diesel fuel is stolen. Eliminating theft reduces actual fuel costs by approximately **fourfold (400%).**

* **Maintenance costs are negligible,** as there is no need to change or top up engine oil, antifreeze, or AdBlue. Oil, fuel, and air filters, batteries, belts, and clutch discs do not require replacement, and brake pad wear is reduced. There are no issues with clogged DPF filters—the replacement of which costs over **3,000 EUR** — nor are there costs for mandatory scheduled servicing or exhaust-related problems during technical inspections.

* There are no routine or major overhauls of the diesel engines. After 10–15 years of intensive two-shift operation, the costs for routine maintenance and major overhauls of the diesel engines amount to **very substantial sums.**

* Full or partial tax exemption, reduced road tolls, and other benefits, depending on the regulations of each individual country.

Operational technical benefits:

* Every new or used ECOES electric truck features an automatic transmission that significantly reduces the electrical and thermal load on the PMSM motor. This extends the driving range and makes operation highly convenient for the driver in demanding urban conditions characterized by frequent stop-and-go traffic.

* ECOES trucks are equipped with a charger that allows for scheduled charging cycles. A solar system with a minimum capacity of 100 kWp is available as an option, providing free electricity for the ECOES truck as well as for other business and household needs.

Why ECOES conversion?

Because the future belongs to electric vehicles. And since this is the future, the main question is **not *whether* but *when*** we will switch to using them. If we do not make the transition to electric vehicles voluntarily and out of conviction, we will do so **under compulsion. This compulsion** stems from the **"stick and carrot"** policy pursued by various international organizations and national governments.

The stick for diesel – Increasingly stringent environmental standards—such as the current Euro 6 and the upcoming Euro 7—are driving stricter emissions regulations and higher prices for mobile diesel machinery. There is also a continuous rise in excise duties and the final retail prices of diesel and gasoline, which are influenced by various market factors, cartel-like dynamics, and regional military conflicts.

The carrot for electrical – Various forms of financial incentives under environmental, "green," and other programs are available for the purchase and use of electric vehicles. It is possible to charge vehicles at home, at a business, or on a farm using reduced off-peak electricity rates. Increasingly extensive and convenient networks of charging stations are being built along road networks. Electricity prices are far more stable than diesel prices, as they rise at a much slower pace and are not subject to market-driven cartel factors or regional military conflicts. **Charging can even be free of charge if one's own solar charging systems are used.**

Under these conditions, the decision to switch from diesel and gasoline to electric power always involves comparing the three types of vehicles: diesel, gasoline, and electric. Diesel and gasoline vehicles perform well, but operating them is becoming increasingly expensive. Electric trucks are much cheaper to run, yet they have a high purchase price and will come to a halt when their battery power runs out.

The connecting link in this transition from diesel and gasoline to electric power is the ECOES conversion. It enables the switch from internal combustion to electric propulsion while simultaneously solving the primary challenge faced by electric vehicles: **maintaining mobility in emergency mode once the battery charge is depleted.**

THE PRIMARY OBJECTIVE OF ECOES JV IS TO OFFER ELECTRIC VEHICLES AT THE PRICE OF THEIR DIESEL COUNTERPARTS THROUGH THE ECOES CONVERSION PROCESS.